

AZURE IAM SECURITY LAB

User, Group, RBAC, Access Testing & Sign-in Logs

Platform: Microsoft Azure + Microsoft Entra ID
Resource Group: RG-IAM-Security-Lab (Australia East)
Test User: IAM Test User -- iam.testuser@BiksecOpsLaboutlook.onmicrosoft.com
Security Group: IAM-Lab-Readers
RBAC Role Assigned: Reader (scoped to RG-IAM-Security-Lab)
MFA: Configured via Microsoft Authenticator
Access Validated: Read: PASS | Delete: FAIL | Create: FAIL
Sign-in Logs: Reviewed via Microsoft Entra ID Monitoring
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1. Lab Overview

This lab demonstrates hands-on experience with Azure Identity and Access Management (IAM) using Microsoft Entra ID. The lab covers the full IAM lifecycle: creating a resource group, provisioning a test user, configuring a security group, assigning RBAC roles, testing access permissions, configuring MFA, and reviewing sign-in logs.

All steps were performed in a live Azure subscription. The Reader role correctly permitted read access while blocking write and delete operations, confirming the principle of least privilege in practice.

2. Lab Objectives

- * Create a Resource Group (RG-IAM-Security-Lab) scoped to Australia East.
- * Provision a test user (IAM Test User) in Microsoft Entra ID.
- * Create a Security Group (IAM-Lab-Readers) and add the test user as a member.
- * Assign the Reader RBAC role to the security group at the resource group scope.
- * Login as the test user and validate: read access works, write/delete fails.
- * Configure MFA using Microsoft Authenticator for the test user account.
- * Review sign-in logs in Microsoft Entra ID Monitoring to confirm sign-in activity.

3. Lab Environment

Platform	Microsoft Azure (Azure Portal)
Identity Service	Microsoft Entra ID (formerly Azure Active Directory)
Subscription	Azure Subscription 1
Resource Group	RG-IAM-Security-Lab
Region	Australia East
Test User	IAM Test User -- iam.testuser@BiksecOpsLaboutlook.onmicrosoft.com
Security Group	IAM-Lab-Readers (Security, Assigned membership)
RBAC Role	Reader (scoped to RG-IAM-Security-Lab)
MFA Method	Microsoft Authenticator App

4. Step 1 -- Create Resource Group

A dedicated Resource Group was created to scope all RBAC assignments for this lab.

Navigation: Azure Portal > Resource groups > Create

Resource Group Name	RG-IAM-Security-Lab
Subscription	Azure Subscription 1
Region	Australia East

Clicked Review + create then Create. Resource group successfully provisioned.

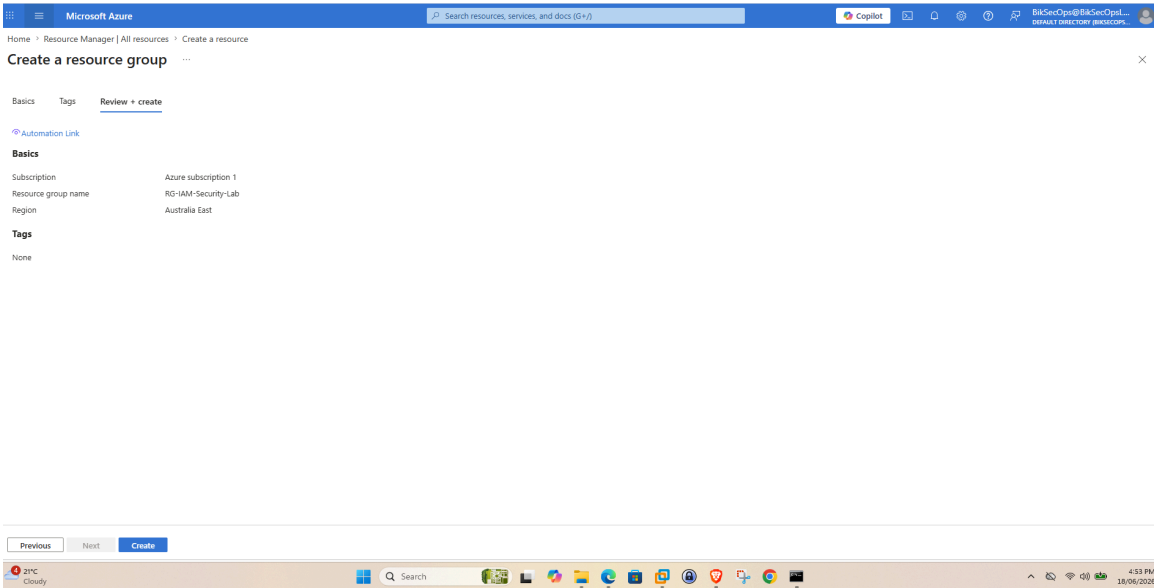


Fig 1 -- Azure Portal: Create resource group Review+create tab -- RG-IAM-Security-Lab, Australia East

5. Step 2 -- Create Test User

A new user was created in Microsoft Entra ID to serve as the test identity for RBAC validation.

Navigation: Microsoft Entra ID > Users > New user > Create new user

Display Name	IAM Test User
User Principal Name	iam.testuser@BiksecOpsLaboutlook.onmicrosoft.com
Password	Auto-generated (temporary)
Account Status	Enabled
User Type	Member

The full username and temporary password were copied for use during access testing.

The screenshot shows the 'Create new user' page in the Microsoft Azure portal. The page is titled 'Create new user' and has a subtitle 'Create a new internal user in your organization'. The 'Basics' tab is selected, and the 'Review + create' button is visible at the bottom. The form contains the following fields and values:

- User principal name: iam.testuser
- Mail nickname: iam.testuser
- Display name: IAM Test User
- Password: Auto-generated (temporary)
- Account enabled: ☒

The domain is set to BiksecOpsLaboutlook.onmicrosoft.com. The 'Derive from user principal name' checkbox is checked for the mail nickname. The 'Auto-generate password' checkbox is checked. The 'Account enabled' checkbox is checked.

Fig 2 -- Entra ID: Create new user -- UPN: iam.testuser, Display name: IAM Test User, Auto-generate password

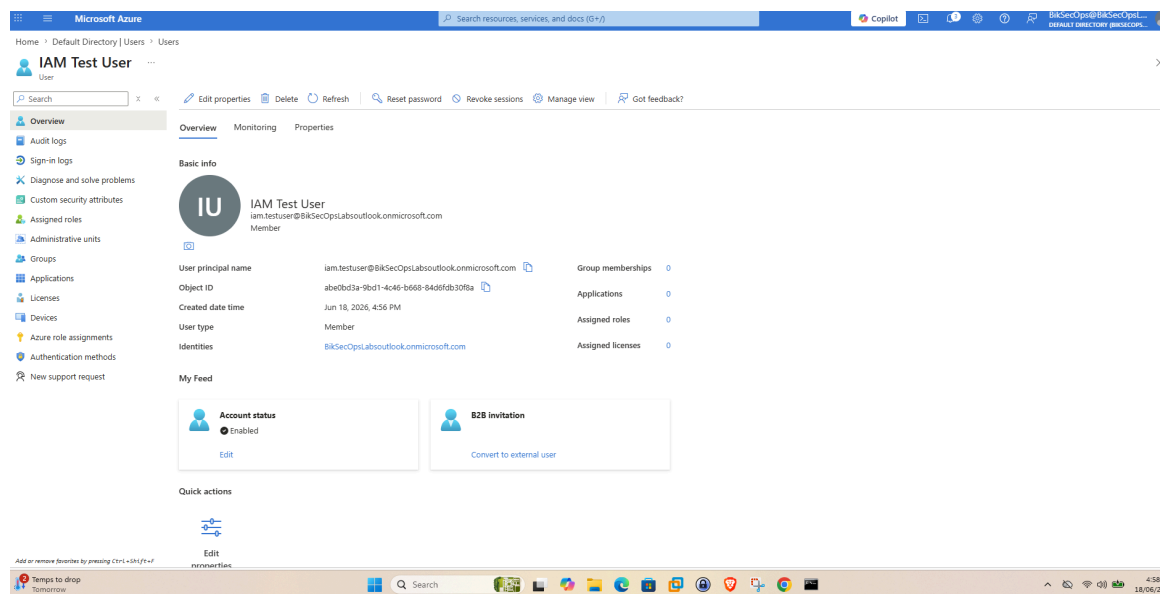


Fig 3 -- Entra ID: IAM Test User profile confirmed -- Account status: Enabled, User type: Member

6. Step 3 -- Create Security Group

A Security Group was created to manage RBAC at group level, following enterprise identity management best practices.

Navigation: Microsoft Entra ID > Groups > New group

Group Type	Security
Group Name	IAM-Lab-Readers
Membership Type	Assigned
Members Added	IAM Test User

After creating the group, IAM Test User was added as a direct member via Members > Add members.

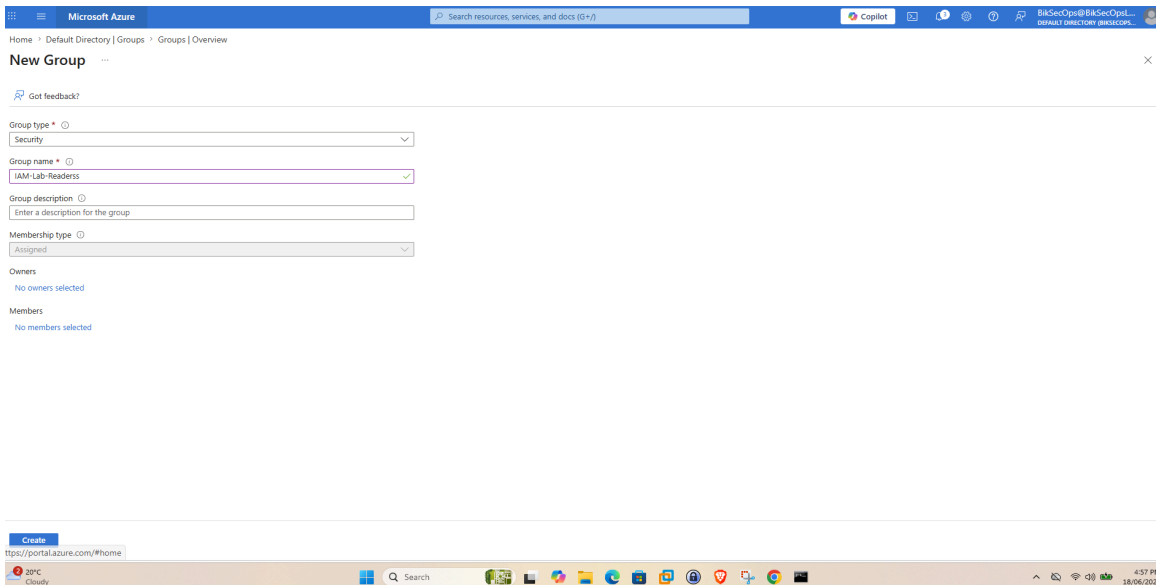


Fig 4 -- Entra ID: New Group form -- Group type: Security, Name: IAM-Lab-Readers, Membership: Assigned

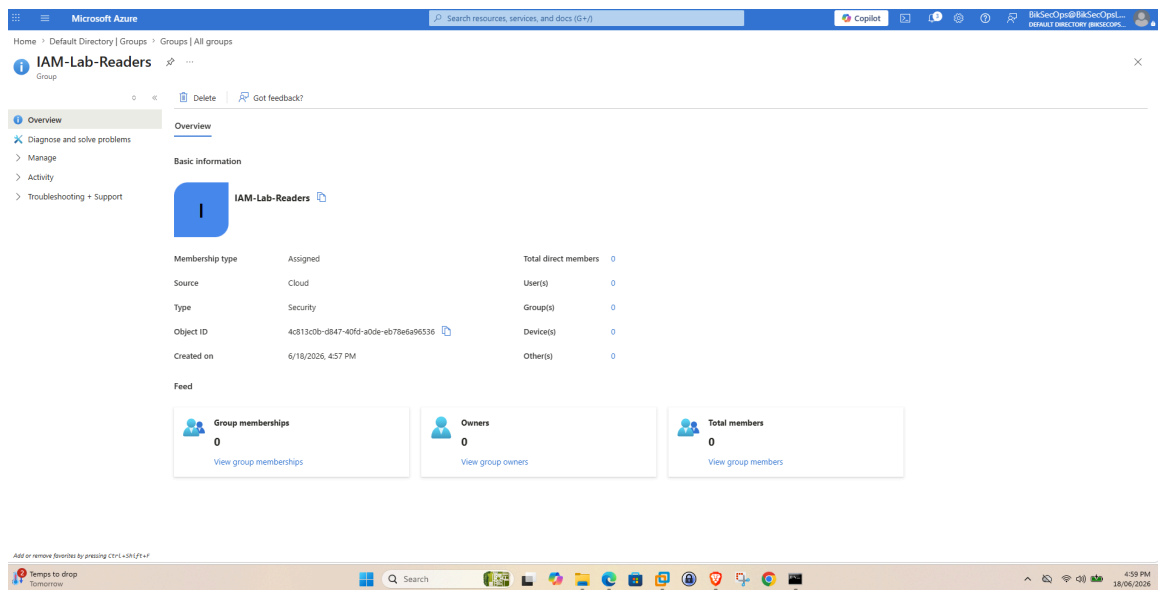


Fig 5 -- Entra ID: IAM-Lab-Readers group overview -- Membership: Assigned, Type: Security, Created 6/10/2026

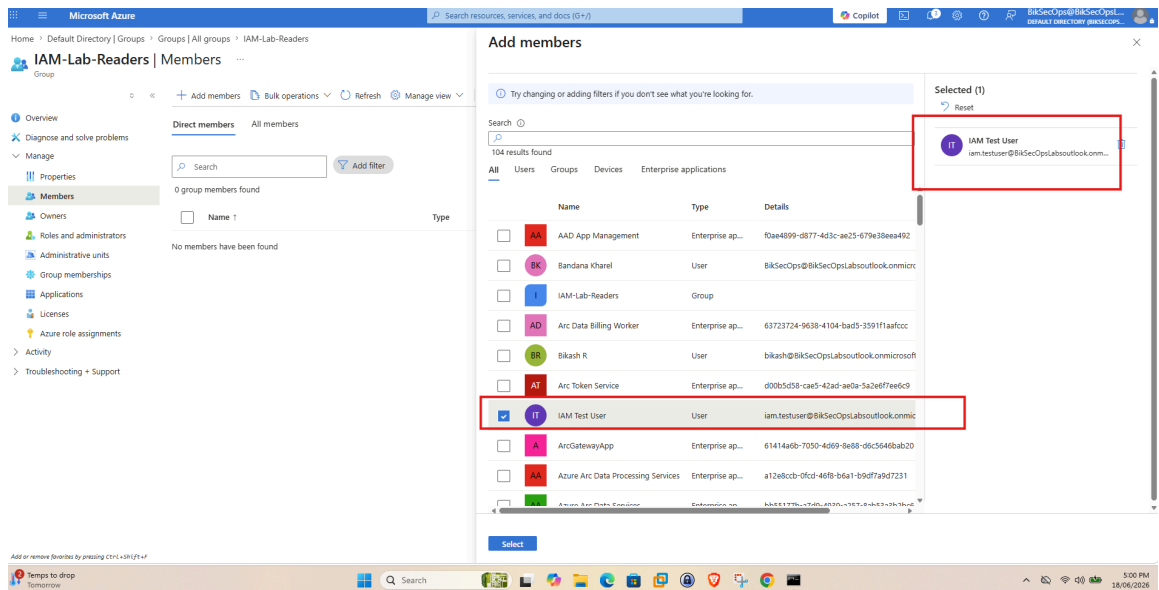


Fig 6 -- Entra ID: Add members -- IAM Test User selected to join IAM-Lab-Readers

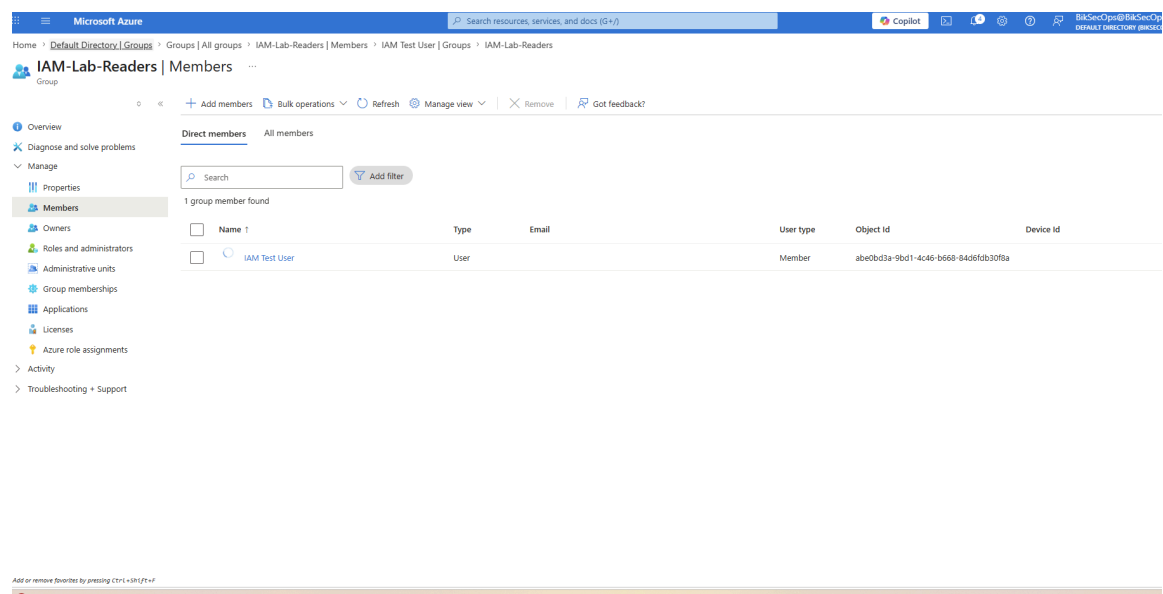


Fig 7 -- Entra ID: IAM-Lab-Readers Members tab -- IAM Test User listed as Direct Member

7. Step 4 -- Assign Reader RBAC Role

The Reader role was assigned to the IAM-Lab-Readers group at the RG-IAM-Security-Lab scope, granting all members read-only access.

Navigation: Resource groups > RG-IAM-Security-Lab > Access control (IAM) > Add > Add role assignment

Role	Reader
Assign Access To	User, group, or service principal
Member (Group)	IAM-Lab-Readers
Scope	RG-IAM-Security-Lab (Resource Group)

IAM-Lab-Readers was selected as the member. Clicked Review + assign to complete the role assignment.

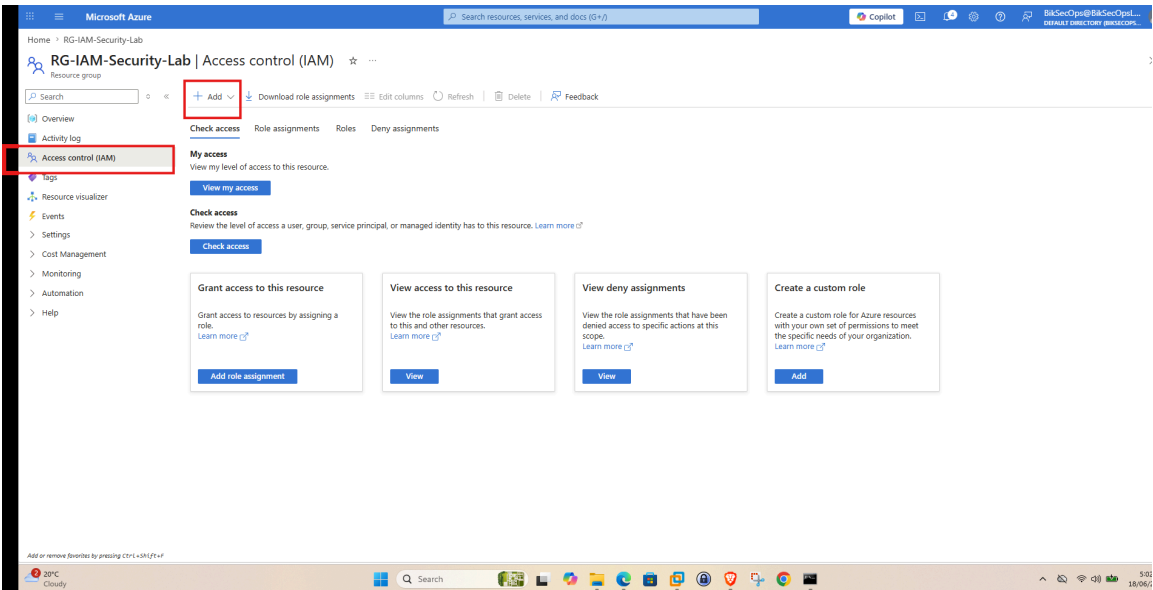


Fig 8 -- Azure Portal: RG-IAM-Security-Lab Access Control (IAM) -- Add role assignment highlighted

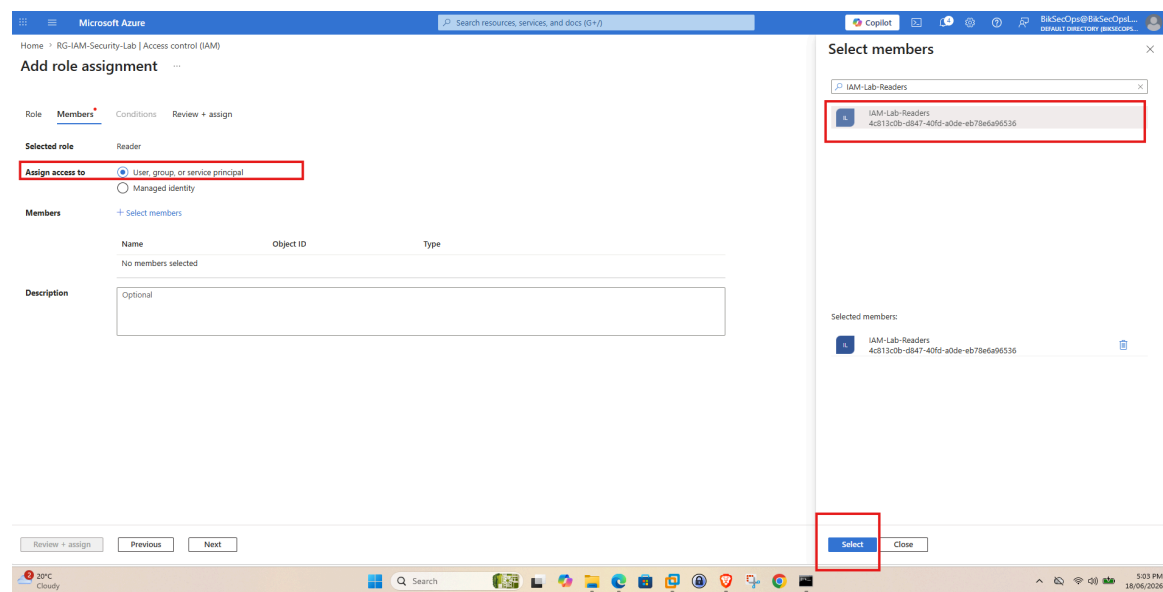


Fig 9 -- Add role assignment Members tab -- IAM-Lab-Readers selected, Assign access to: User/group/service principal

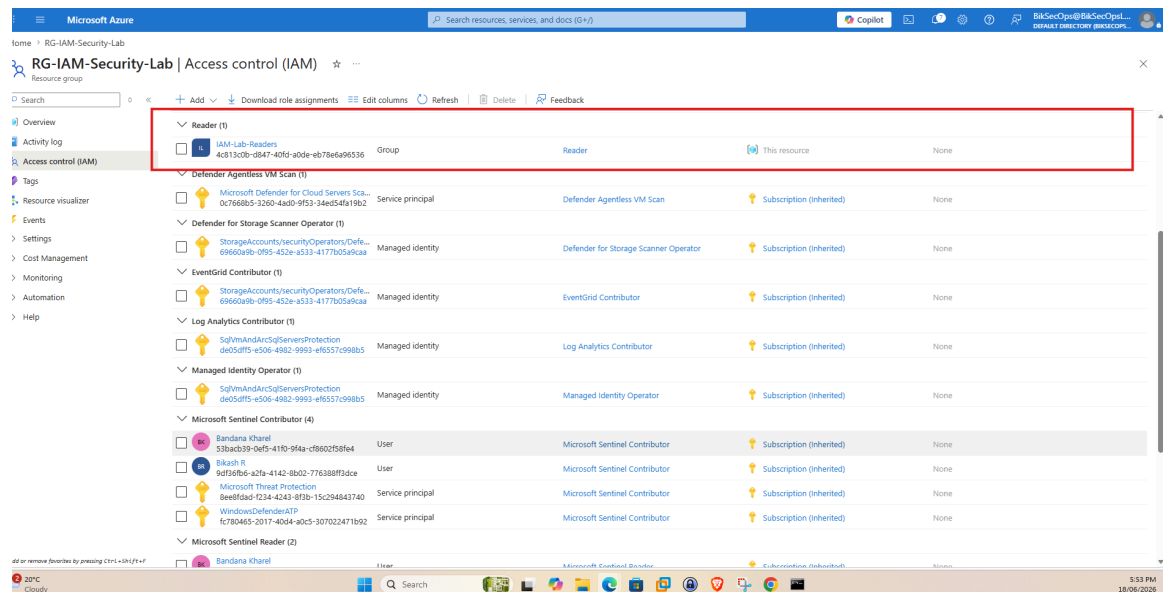


Fig 10 -- RG-IAM-Security-Lab IAM Role assignments -- Reader role assigned to IAM-Lab-Readers confirmed

8. Step 5 -- Login as Test User & Access Testing

The IAM Test User account was used to log in via a private browser to validate that the Reader role was correctly enforced.

Login Process

- * Opened Incognito/Private browser window
- * Navigated to <https://portal.azure.com>
- * Logged in with iam.testuser credentials and temporary password
- * Changed password when prompted
- * Configured MFA using Microsoft Authenticator app

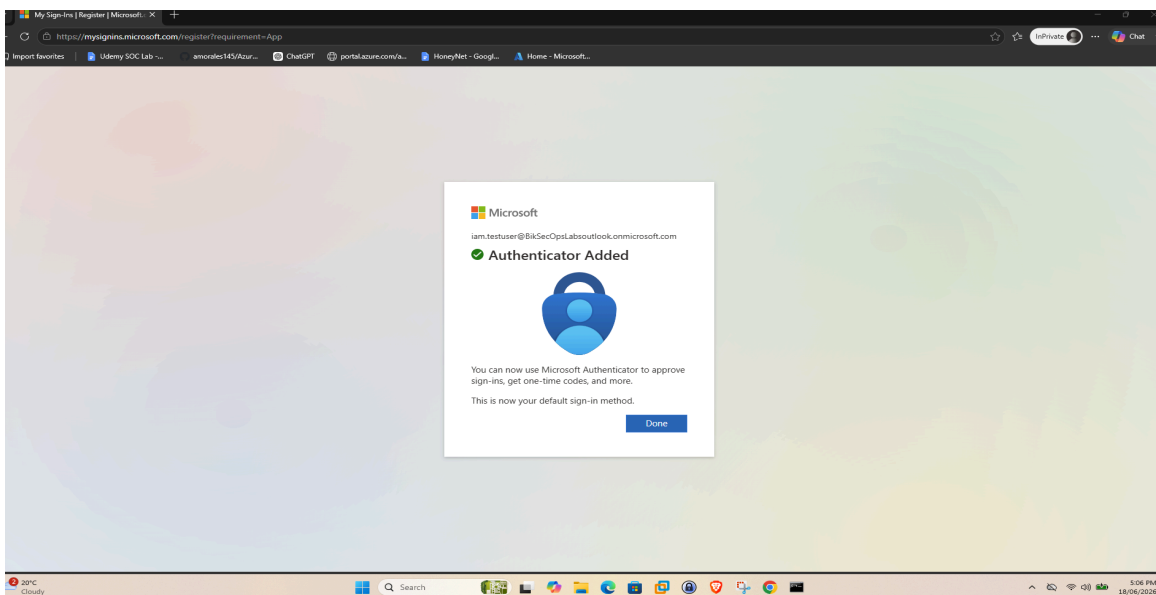


Fig 11 -- Microsoft Authenticator setup complete: "Authenticator Added" for iam.testuser@BiksecOpsLaboutlook.onmicrosoft.com

Access Test Results

Test Action	Expected	Result
View RG-IAM-Security-Lab	Permitted (Reader)	PASS
Delete Resource Group	Denied -- AuthorizationFailed	FAIL (as expected)
Create new Resource Group	Denied -- no permissions	FAIL (as expected)

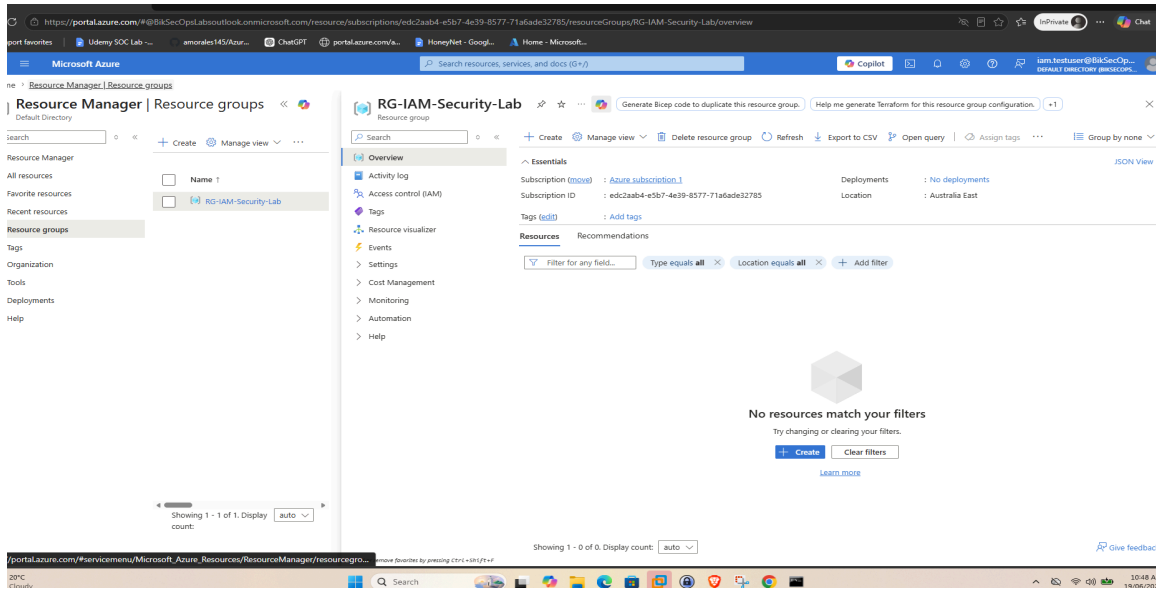


Fig 12 -- Test User Portal: RG-IAM-Security-Lab visible and accessible -- Reader role WORKING

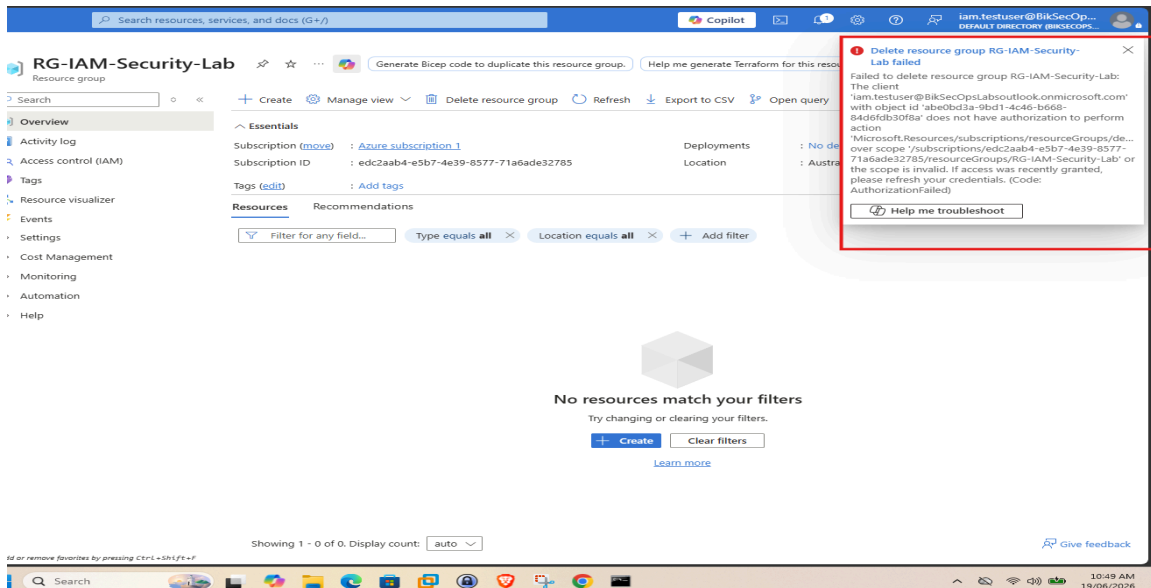


Fig 13 -- Delete resource group FAILED -- AuthorizationFailed: no authorization to delete (Reader cannot delete)

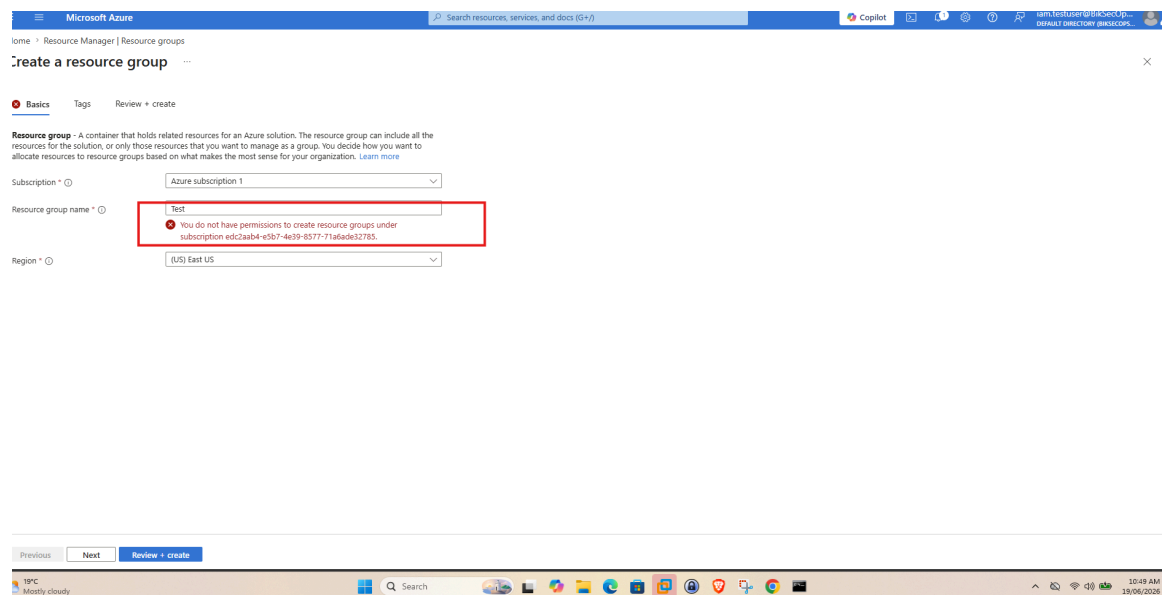


Fig 14 -- Create new resource group FAILED -- "You do not have permissions to create resource groups"

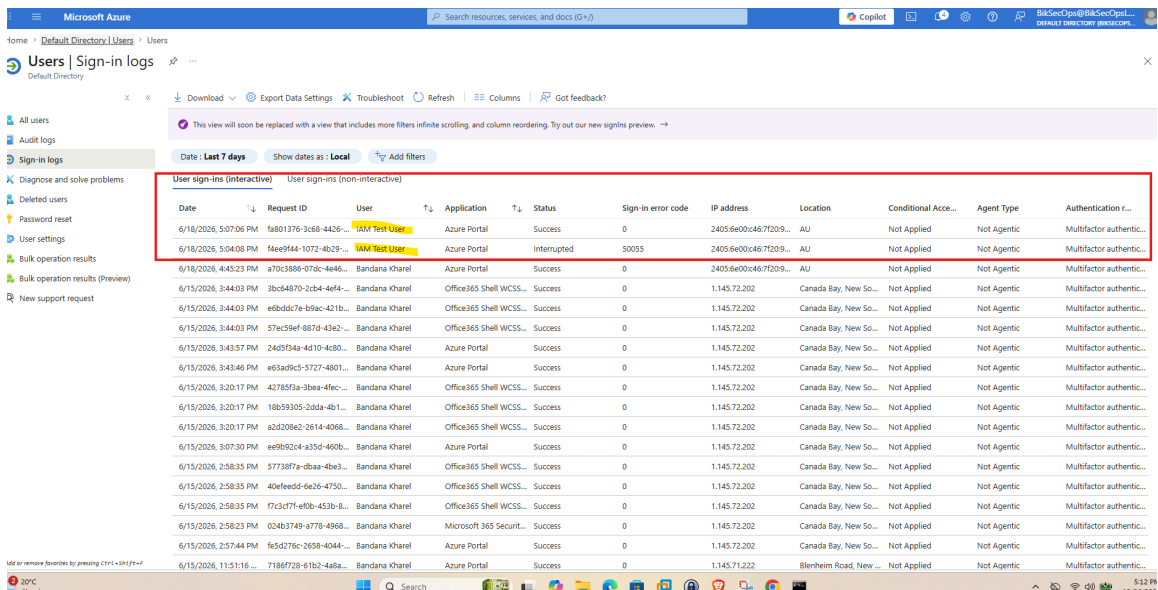
9. Step 6 -- Review Sign-in Logs

After access testing, the admin account was used to review Entra ID Sign-in logs to confirm the test user sign-in activity.

Navigation: Microsoft Entra ID > Monitoring & health > Sign-in logs

Filter Applied	User: IAM Test User
Date Range	Last 7 days
Application	Azure Portal
Sign-in Status	Success and Interrupted (MFA setup)
Authentication Method	Multifactor Authentication
Location	AU (Australia)
IP Address	2409:8a08:c46:7029:...

Sign-in log confirmed two entries for IAM Test User: one Success and one Interrupted (MFA registration), both from Azure Portal, Australia.



User sign-ins (interactive)											
Date	Request ID	User	Application	Status	Sign-in error code	IP address	Location	Conditional Ace...	Agent Type	Authentication c...	
6/18/2026, 5:07:06 PM	fa801376-3c68-4426-...	IAM Test User	Azure Portal	Success	0	2405:6e00:c46:7f209...	AU	Not Applied	Not Agentic	Multifactor authentic...	
6/18/2026, 5:04:08 PM	f4ee5944-1072-4b29-...	IAM Test User	Azure Portal	Interrupted	50055	2405:6e00:c46:7f209...	AU	Not Applied	Not Agentic	Multifactor authentic...	
6/18/2026, 4:45:23 PM	a70c3886-07d6-4e46...	Bandana Kharel	Azure Portal	Success	0	2405:6e00:c46:7f209...	AU	Not Applied	Not Agentic	Multifactor authentic...	
6/15/2026, 3:44:03 PM	3bc54870-2cb4-4ef4...	Bandana Kharel	Office365 Shell WCSS...	Success	0	1.145.72.202	Canada Bay, New So...	Not Applied	Not Agentic	Multifactor authentic...	
6/15/2026, 3:44:03 PM	e6bdc7e-b9ac-421b...	Bandana Kharel	Office365 Shell WCSS...	Success	0	1.145.72.202	Canada Bay, New So...	Not Applied	Not Agentic	Multifactor authentic...	
6/15/2026, 3:44:03 PM	57ec59ef-887d-43a2...	Bandana Kharel	Office365 Shell WCSS...	Success	0	1.145.72.202	Canada Bay, New So...	Not Applied	Not Agentic	Multifactor authentic...	
6/15/2026, 3:43:57 PM	24d5f34a-4d10-4c80...	Bandana Kharel	Azure Portal	Success	0	1.145.72.202	Canada Bay, New So...	Not Applied	Not Agentic	Multifactor authentic...	
6/15/2026, 3:43:46 PM	e63a9dc5-5727-4801...	Bandana Kharel	Azure Portal	Success	0	1.145.72.202	Canada Bay, New So...	Not Applied	Not Agentic	Multifactor authentic...	
6/15/2026, 3:20:17 PM	4278f53a-3bea-4fec...	Bandana Kharel	Office365 Shell WCSS...	Success	0	1.145.72.202	Canada Bay, New So...	Not Applied	Not Agentic	Multifactor authentic...	
6/15/2026, 3:20:17 PM	18b59305-2dda-4b1...	Bandana Kharel	Office365 Shell WCSS...	Success	0	1.145.72.202	Canada Bay, New So...	Not Applied	Not Agentic	Multifactor authentic...	
6/15/2026, 3:20:17 PM	a2d208e2-2614-4068...	Bandana Kharel	Office365 Shell WCSS...	Success	0	1.145.72.202	Canada Bay, New So...	Not Applied	Not Agentic	Multifactor authentic...	
6/15/2026, 3:07:30 PM	ee9b92c4-a35d-460b...	Bandana Kharel	Azure Portal	Success	0	1.145.72.202	Canada Bay, New So...	Not Applied	Not Agentic	Multifactor authentic...	
6/15/2026, 2:58:35 PM	5773877a-dbaa-4be3...	Bandana Kharel	Office365 Shell WCSS...	Success	0	1.145.72.202	Canada Bay, New So...	Not Applied	Not Agentic	Multifactor authentic...	
6/15/2026, 2:58:35 PM	40efeedd-6a26-4750...	Bandana Kharel	Office365 Shell WCSS...	Success	0	1.145.72.202	Canada Bay, New So...	Not Applied	Not Agentic	Multifactor authentic...	
6/15/2026, 2:58:35 PM	f7c3d7f1-ef0b-453b...	Bandana Kharel	Office365 Shell WCSS...	Success	0	1.145.72.202	Canada Bay, New So...	Not Applied	Not Agentic	Multifactor authentic...	
6/15/2026, 2:58:23 PM	024b3749-4778-4966...	Bandana Kharel	Microsoft 365 Secur...	Success	0	1.145.72.202	Canada Bay, New So...	Not Applied	Not Agentic	Multifactor authentic...	
6/15/2026, 2:57:44 PM	fa5d276c-2658-4044...	Bandana Kharel	Azure Portal	Success	0	1.145.72.202	Canada Bay, New So...	Not Applied	Not Agentic	Multifactor authentic...	
6/15/2026, 11:51:16 ...	7186f728-61b2-4a8a...	Bandana Kharel	Azure Portal	Success	0	1.145.71.222	Blenheim Road, New ...	Not Applied	Not Agentic	Multifactor authentic...	

Fig 15 -- Entra ID Sign-in Logs: IAM Test User highlighted -- Success & Interrupted entries, Azure Portal, AU location, MFA auth

10. Skills Demonstrated

Microsoft Entra ID User Provisioning	Azure Resource Group Management	RBAC Role Assignment
Security Group Management	MFA Configuration (Authenticator)	Access Testing & Validation
Azure IAM (Access Control)	Sign-in Log Review & Monitoring	Principle of Least Privilege
Azure Portal Navigation	Identity Lifecycle Management	Cloud Identity & Access Security

11. Conclusion

This lab successfully demonstrated the complete Azure IAM lifecycle using Microsoft Entra ID and Azure RBAC. A test user was provisioned, added to a security group, and granted Reader-level access to a scoped resource group. Access testing confirmed the Reader role correctly permits read operations while blocking delete and create actions, validating the principle of least privilege in a live Azure environment.

MFA was configured for the test user using Microsoft Authenticator, and all sign-in activity was confirmed in the Entra ID Sign-in logs. This lab reflects real-world identity management practices used in enterprise Azure environments.

References

- Microsoft Azure Portal -- <https://portal.azure.com>
- Microsoft Entra ID -- <https://entra.microsoft.com>
- Azure RBAC Documentation -- <https://learn.microsoft.com/azure/role-based-access-control/>
- Microsoft Authenticator -- <https://www.microsoft.com/security/mobile-authenticator-app>